

Test Case Matrix				
ID	REST API	Test Cases	Assertion / Oracle	Authority
ACC-1	POST /accounts	Create valid account	Validate: - 201 Created - account response schema - returned account_id, document_number valid	Contract
ACC-2	POST /accounts	Minimum document length	Validate: - 201 Created - account response schema - returned account_id, document_number valid	Hearsay + Assumption
ACC-3	POST /accounts	Maximum document length	Validate: - 201 Created - account response schema - returned account_id, document_number valid	Hearsay + Assumption
ACC-4	POST /accounts	Empty document_number	Validate: - 400 Bad Request - error: document_number is required	Hearsay + Assumption
ACC-5	POST /accounts	Below minimum document length	Validate: - 422 Unprocessable Entity - error: document_number should be b/w 10 and 14 digits	Hearsay + Assumption
ACC-6	POST /accounts	Above maximum document length	Validate: - 422 Unprocessable Entity - error: document_number should be b/w 10 and 14 digits	Hearsay + Assumption
ACC-7	POST /accounts	Duplicate document_number	Validate: - 409 Conflict - error: document_number already exists	Domain Reasoning + Assumption
ACC-8	POST /accounts	Invalid document characters	Validate: - 422 Unprocessable Entity - error: document_number should only contain digits	Hearsay + Assumption
ACC-9	GET /accounts/{accountId}	Retrieve created account	Validate: - 200 OK - account response schema - ID and document_number consistent	Contract
ACC-10	GET /accounts/{accountId}	Fetch unknown account	Validate: - 404 Not Found - error: account not found	Contract
TXN-1	POST /transactions	Normal Purchase	Validate: - 201 Created - transaction response schema - account_id consistent with request - amount consistent with request with -ve sign - event_date in correct format - operation_type_id is 1 - transaction_id valid - type is "debit"	Contract + Hearsay + Domain

TXN-2	POST /transactions	Installment Purchase	Validate: - 201 Created - transaction response schema - account_id consistent with request - amount consistent with request with -ve sign - event_date in correct format - operation_type_id is 2 - transaction_id valid - type is "debit"	Contract + Hearsay + Domain + Assumption
TXN-3	POST /transactions	Withdrawal	Validate: - 201 Created - transaction response schema - account_id consistent with request - amount consistent with request with -ve sign - event_date in correct format - operation_type_id is 3 - transaction_id valid - type is "debit"	Contract + Hearsay + Domain + Assumption
TXN-4	POST /transactions	Credit Voucher	Validate: - 201 Created - transaction response schema - account_id consistent with request - amount consistent with request with +ve sign - event_date in correct format - operation_type_id is 4 - transaction_id valid - type is "credit"	Contract + Hearsay + Domain + Assumption
TXN-5	POST /transactions	Amount passed as integer	Validate: - 201 Created - returned amount equals caller amount.	Contract + Hearsay
TXN-6	POST /transactions	Amount passed as decimal	Validate: - 201 Created - returned amount equals caller amount.	Contract + Assumption
TXN-7	POST /transactions	Zero amount	Validate: - 422 Unprocessable Entity - error: amount can't be zero	Hearsay + Domain Reasoning
TXN-8	POST /transactions	Negative amount	Validate: - 422 Unprocessable Entity - error: amount can't be negative	Hearsay
TXN-9	POST /transactions	Very large amount - max / max + 1	Validate boundary/overflow behavior.	Domain Reasoning + Assumption
TXN-10	POST /transactions	High precision amount, e.g. 10.123456	Validate - 201 Created - rounding behavior.	Domain Reasoning + Assumption
TXN-11	POST /transactions	Sub-unit amount, e.g. Rs 0.01 (1 paisa)	Validate - 201 Created - amount stored in smallest unit (requires state oracle).	Hearsay
TXN-12	POST /transactions	Unknown operation_type_id	Validate: - 422 Unprocessable Entity - error: possible operation type - 1, 2, 3, 4	Domain Reasoning

TXN-13	POST /transactions	Non existent account	Validate: - 422 Unprocessable Entity - error: possible operation type - 1, 2, 3, 4	Domain Reasoning + Assumption
TXN-14	POST /transactions	Missing required field - Omit account_id, amount, operation_type_id individually.	Validate: - 400 Bad Request - error: account_id, amount, and operation_type_id are required	Domain Reasoning + Assumption
TXN-15	POST /transactions	Wrong field type - Wrong JSON types/null for required fields.	Validate: - 400 Bad Request - error: malformed request	Assumption
TXN-16	POST /transactions	Replay: same idempotency key + same body	Validate: - 200 OK - original result on reply - exactly one transaction (requires state oracle)	Hearsay + Assumption
TXN-17	POST /transactions	Replay: same idempotency key + different body	Validate: - 409 Conflict	Domain Reasoning + Assumption
TXN-18	POST /transactions	Concurrent same idempotency key replay	Validate N concurrent requests: - create exactly one transaction (requires state oracle). - HTTP status code based on same body or different body	Hearsay + Domain Reasoning + Assumption
TXN-19	POST /transactions	Missing idempotency key	Validate: - 400 Bad Request - error: idempotency key is required	Hearsay + Assumption
CMN-1	POST /accounts GET /accounts/{accountid} POST /transactions	Request-ID success	Validate header exists and matches documented format.	Hearsay
CMN-2	POST /accounts GET /accounts/{accountid} POST /transactions	Request-ID error	Validate header on errors and matches documented format.	Hearsay
CMN-3	POST /accounts GET /accounts/{accountid} POST /transactions	Method not allowed	Validate: - 405 Method Not Allowed.	Contract
CMN-4	POST /accounts GET /accounts/{accountid} POST /transactions	Malformed JSON	Validate: - 400 Bad Request - error: malformed request	Contract
CMN-5	POST /accounts GET /accounts/{accountid} POST /transactions	Wrong content type - send unsupported media type	Validate: - 415 Unsupported Media Type	Contract + Assumption
CMN-6	POST /accounts GET /accounts/{accountid} POST /transactions	Empty body	Validate: - 400 Bad Request - error: <field_names> are required	Contract + Domain + Assumption
CMN-7	POST /accounts GET /accounts/{accountid} POST /transactions	Oversized body	Validate: - 400 Bad Request - error: malformed request	Assumption
CMN-8	POST /accounts GET /accounts/{accountid} POST /transactions	Response content type	Validate: - content-type is application/json.	Contract

Amount Edge Case Matrix			
Test Case	Example	Expected	Actual (as per contract)
Zero	0	Validate: - 422 Unprocessable Entity - error: amount can't be zero	Zero amount is schema-valid.
Negative	-1	Validate: - 422 Unprocessable Entity - error: amount can't be negative	Negative amount is schema-valid.
Very Large	max / max + 1	Validate boundary/overflow behavior.	Contract doesn't enforce and minimum and maximum value.
High Precision	10.123456	Validate - 201 Created - rounding behavior.	Contract doesn't enforce any precision rule.
Sub Unit	smallest fraction, e.g. Rs 0.01 (1 paisa)	Validate - 201 Created - amount stored in smallest unit.	There is no currency field or unit declaration anywhere in the supplied contract.

Traceability Matrix			
Test Case	C1 Risk Retired	Defect Caught	Gate
ACC-1, ACC-7	R9 - account identity duplicated	Two accounts created for one document number when uniqueness is required.	PR
TXN-1, TXN-2, TXN-3, TXN-4	R2 - incorrect transaction size/type	Operation mapped to wrong financial direction/type	PR
TXN-5, TXN-6, TXN-7, TXN-8, TXN-9, TXN-11	R3 - amount/unit/precision	Amount recorded incorrectly, causes customer balances and financial records to be inaccurate.	PR + Nightly
TXN-12	R7 - Unsupported operation type reaches a financial transaction	An invalid operation can change money in an incorrect way.	PR
ACC-10, TXN-13	R6 - transaction created for an invalid/non-existent account	Transaction accepted for missing account	PR
TXN-16, TXN-18, TXN-19	R1 - duplicate transaction during retry/concurrency	Double debit/credit	Nightly + Prerelease
TXN-17	R5 - same idempotency key reused with a different payload	A retry key can become associated with two different financial operations.	Nightly + Prerelease

CMN-1, CMN-2	R10 - missing/incorrect request correlation ID	Trace correlation absent/invalid	PR	
ACC-2, ACC-3, ACC-4, ACC-5, ACC-6, ACC-8, TXN-15, CMN-3, CMN-4, CMN-5, CMN-6, CMN-7, CMN-8	R11/R13 - HTTP/schema/request handling	Unexpected method/media/parser/body behavior	PR + Nightly	
Contract Tests, TXN-14	R11 - consumer breakage	Provider changes field/type/status/media type and breaks consumer	PR	
Audit verification	R4 - audit divergence	Business write commits without matching audit record	Prerelease	
Transaction state verification, ACC-9	R8 - lost transaction	Success response does not correspond to durable transaction	Prerelease	